

Review

Formation of Multispecies Biofilms and Their Resistance to Disinfectants in Food Processing Environments: A Review

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ABSTRACT

In food processing environments, various microorganisms can adhere and aggregate on the surface of equipment, resulting in the formation of multispecies biofilms. Complex interactions among microorganisms may affect the formation of multispecies biofilms and resistance to disinfectants, which are food safety and quality concerns. This article reviews the various interactions among microorganisms in multispecies biofilms, including competitive, cooperative, and neutral interactions. Then, the preliminary mechanisms underlying the formation of multispecies biofilms are discussed in relation to factors, such as quorum-sensing signal molecules, extracellular polymeric substances, and biofilm-regulated genes. Finally, the resistance mechanisms of common contaminating microorganisms to disinfectants in food processing environments are also summarized. This review is expected to facilitate a better understanding of interspecies interactions and provide some implications for the control of multispecies biofilms in food processing.

HIGHLIGHTS

- Multispecies interaction of biofilms is often cooperative, competitive, or neutral.
- QS molecules, EPS, and biofilm-regulated genes affect multispecies biofilm formation.
- Disinfectant resistance of multispecies biofilms depends on EPS and biofilm age.
- Studies of multispecies biofilms are less focused on high-diversity biofilms.

Key words: Disinfectants; Food processing environments; Interactions; Mechanisms; Multispecies biofilms

To adapt to various stress conditions imposed by the environment, microorganisms have developed different survival strategies, including biofilm formation. Biofilms are a community of microorganisms encapsulated in extracellular polymeric substances (EPS), mainly composed of polysaccharides, proteins, lipids, and nucleic acids (15). About 40 to 80% of bacteria on the Earth can form biofilms, just as some fungi (22, 27, 37). However, most biofilms found in natural, clinical, or industrial environments are formed by two or more species of microorganisms instead of a single species and are, thus, called multispecies biofilms (25).

Due to the species diversity present in the raw materials, multispecies biofilms are often formed on food contact surfaces (e.g., cutters, spiral conveyors, and related equipment) and nonfood contact surfaces (e.g., drains and pipes) in the meat and dairy industry (75, 103). Even after normal hygiene and disinfection procedures, several microorganisms can still remain in food production

environments, such as *Staphylococcus*, *Stenotrophomonas*, *Streptococcus*, *Pseudomonas*, *Acinetobacter*, and *Listeria* (24, 34, 53, 65). These microorganisms may affect food quality and safety, because they not only coexist in biofilms but also persist in production environments, posing continual and potential contamination risks during food manufacture.

Microscopic techniques have helped reveal three types of bacterial organization in multispecies biofilms: microcolonies, which form separate small niches in parallel; layered structures, in which each species is located in a different layer; and multispecies mixing patterns (35, 61, 74). The distribution of these structures largely depends on the type of microorganisms and leads to competitive and cooperative or neutral interactions (3). These interactions bring about some ecological advantages for microorganisms, such as the availability of nutrients and resistance to harsh environments and disinfectants. Biofilm formation enhances the persistence of internal microorganisms inside the biofilm matrix. This persistence leads to an increase of microbial load in the food processing environment and in the subsequent food products, causing food spoilage,

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shortened shelf life, and increased risk of infectious diseases originating from food sources (16). For example, in the food industry, pathogenic bacteria can form biofilms in processing facilities, causing corrosion of facility surfaces and blocking of pipes (28). *Aeromonas* and *Pseudomonas* are important bacteria associated with the spoilage of refrigerated freshwater fish. The coculture of these two bacteria could significantly increase the maximum growth rate, promote biofilm formation, and improve the spoilage capacity of bacterial strains, resulting in spoilage of grass carp (113).

Competition, cooperation, and communication of microorganisms in food processing environments lead to the complex behavior of multispecies biofilms, enhancing the infectivity and resistance of microorganisms in the biofilms (93). An increase in disinfectant resistance will significantly reduce the efficiency of cleaning and disinfection processes, allowing microorganisms to remain in food and food processing equipment (97). Microorganisms that form multispecies biofilms may be affected by other microorganisms, thus changing the survival status as a potential source of contamination. The multispecies biofilms formed by *Lactobacillus plantarum* and *Listeria innocua* can change from an almost completely unculturable state to a culturable state, representing a potential source of contamination (76).

Therefore, the formation of multispecies biofilms can give rise to the spread and persistence of harmful microorganisms in food processing environments, shorten the shelf life of food, and increase the probability of human illness after ingestion of contaminated food. Therefore, special research attention should be paid to the various potential negative effects of multispecies biofilms. This article will discuss these aspects from three perspectives: the preliminary mechanisms for the formation of multispecies biofilms, the interactions among the microorganisms in multispecies biofilms, and the preliminary mechanisms for the resistance of microorganisms in multispecies biofilm to disinfectants. From the perspective of food safety, this review will contribute to more accurate risk analysis and the development of effective strategies for the prevention and control of harmful multispecies biofilms.

PRELIMINARY MECHANISMS FOR THE FORMATION OF MULTISPECIES BIOFILMS

Research efforts have revealed that the formation process of multispecies biofilms consists of five successive steps: (i) reversible attachment; (ii) irreversible attachment; (iii) microcolony formation; (iv) biofilm maturation; and (v) dispersion or separation (Fig. 1). The formation of bacterial biofilms starts with the formation of an appropriate surface layer after the absorption of organic or inorganic molecules (such as polysaccharides, proteins, lipids, and fatty acids) and then single or mixed communities are embedded into heterostructured EPS (16). Once bacteria attach to biological or abiotic surfaces, they communicate with each other by using an extracellular signaling system on the basis of quorum sensing (QS) (71). In the whole formation process of multispecies biofilms, biofilm-regulated genes will be activated and play corresponding roles. In addition, biofilm

formation is a process in which microorganisms change phenotypes for adaptation to environmental stress or immune response. Multispecies interactions increase the probability of changes in biofilm-regulated genes. Therefore, the formation mechanisms of multispecies biofilms are closely related to QS, EPS, biofilm-regulated genes, and other factors (Fig. 2).

QS signal molecules. QS is a way in which species sense cell density and, accordingly, regulate gene expression (7). There has been increasing evidence showing that multiple microorganisms can sense specific QS signal molecules, enabling them to perceive and react with other microorganisms nearby (20). QS plays a key role in the collective behavior of bacteria, including the initial stage of biofilm attachment and the maturation or diffusion of biofilms at the intraspecies and interspecies levels (87). QS molecules produced by bacteria are called autoinducers (AI), used for one-way, two-way, or multiway communication (85). When the concentration of AI reaches the threshold value, the expression of certain genes will be changed (108). In general, AI are involved in biofilm formation and microorganism motility and also coordinate intra- and interspecies interactions (32, 48). Although fungi have not been shown to produce the analogs of bacterial AI, QS signal molecules, such as farnesol, tyrosol, phenylethanol, and tryptophol, have been identified in fungi (Fig. 2A) (110). Currently, the importance of some QS signal molecules in multispecies biofilms has been gradually revealed, such as acyl-homoserine lactones (AHL) and autoinducer-2 (AI-2; Fig. 2A).

AHL-mediated QS is a vital regulatory mechanism for the formation and structure of gram-negative bacterial biofilms, regulating the maturation of biofilms (52). Chandler et al. (12) reported that AHL is indispensable for interspecies competition. Zhu et al. (114) found that there are competitive interactions between *Pseudomonas fluorescens* and *Shewanella baltica*, and *S. baltica* may inhibit the production of AHL in dual-species biofilms or consume the AHL secreted by *P. fluorescens*. de Almeida et al. (17) revealed that *Salmonella* Enteritidis can form stronger biofilms in the presence of AHL from other species.

AI-2, a class of cyclic oligopeptides, is a nonspecific QS signal molecules produced by both gram-negative and gram-positive bacteria (98). It has been proved that S-ribosyl homocysteine is a precursor of AI-2, playing an indispensable role in the biofilm formation of *Listeria monocytogenes* (8). Similarly, the QS system of *Staphylococcus aureus* seems to be involved in the regulation of *L. monocytogenes* attachment and biofilm formation in an early stage (91). Laganenka and Sourjik (51) have established the importance of the AI-2-mediated signal in mixed-species biofilm formation of *Enterococcus faecalis* and *Escherichia coli*; *lsr* encodes the LSR protein. AI-2 accumulates in the extracellular medium with early growth, and once a certain cell density is reached, these levels are depleted by import via the Lsr transport system. In the dual-species biofilms, *E. faecalis* could secrete extra AI-2 to enhance the expression of *E. coli lsr*, which promotes the

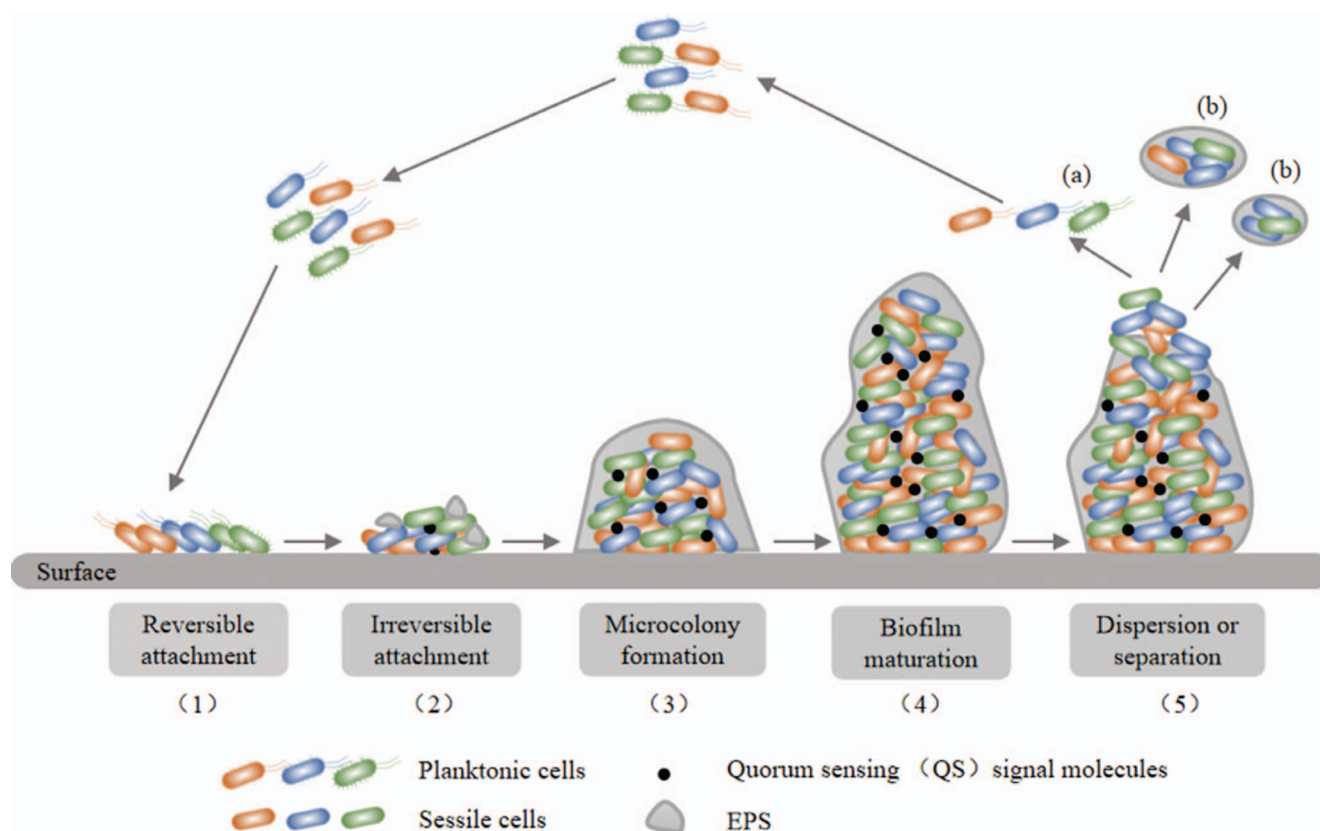


FIGURE 1. Hypothetical processes of multispecies biofilm formation: (1) reversible attachment, planktonic cells attach nonspecifically to the surface and aggregate loosely, which can dissociate and revert to the form of planktonic cells; (2) irreversible attachment, microbial cells irreversibly attach to the surface, the cells begin to secrete EPS and QS factors, and the aggregated population continues to grow; (3) microcolony formation, microbial cells interact to form microcolonies, and both interspecies and intraspecies cells communicate through QS factors and EPS; (4) biofilm maturation, microcolonies continue to grow and form mature biofilm structures; (5) dispersion or separation: (a) the microbial cells diffuse actively and return to the planktonic states; (b) changes in environmental conditions, such as cleaning and disinfection measures, passively detach biofilm aggregates from mature biofilms (50, 70). The spatial arrangement of microorganisms in the multispecies biofilms in this figure is based on the mixed model (61).

coaggregation of *E. coli* and *E. faecalis*, increasing the stress resistance of both *E. coli* and *E. faecalis*. Thus, another putative function for the Lsr is interference in AI-2-dependent QS. The interaction between *P. fluorescens* and *S. aureus* in dual-species biofilms could promote the production of extracellular polysaccharides regulated by AI-2 and result in the complex biofilm structures (107). *Pseudomonas aeruginosa* and *P. fluorescens* were found to significantly inhibit the biofilm formation of *Aeromonas hydrophila* on crab coupons and significantly reduce the levels of AHL and AI-2, supporting the relationship between QS and biofilm formation (43).

QS is an indispensable mechanism on the basis of the production, secretion, and subsequent concentration-dependent reactions of the signal molecule AI (41). Interference with QS can help inhibit the formation of biofilms by foodborne pathogens and reduce possible disease outbreaks. In the future, it will be necessary to study QS in the coculture of multiple microorganisms.

EPS. EPS are composed of a series of polymers, such as carbohydrates, proteins, extracellular DNA, and extracellular RNA, which interact with and transport small

molecules such as QS signals and redox shuttles to coordinate the activity of the whole biofilm matrix (90). The existence of EPS not only contributes to the establishment of biofilms but also promotes growth, including mediating cell attachment and expanding the volume of cell clusters (46, 68). The expansion of cell cluster volume allows the entry of EPS-producing strains into the nutrient-rich area in the biofilms, while those strains that do not produce EPS are at a disadvantage. EPS-producing strains and non-EPS-producing strains will compete for nutrients and spaces, with the former generally having a stronger biofilm formation ability, and thus a higher possibility to gain a foothold in multispecies biofilms (106). In the formation of biofilms by two or more microbial strains, the interactions among different EPS components may change the overall composition of EPS.

EPS of microbial communities can be divided into two categories: (i) the components related to the cell surface, such as fimbriae, flagella, and functional amyloids that regulate bacterial attachment, stability, and autoimmune response and (ii) extracellular secretion, such as bacterial extracellular polysaccharides, proteins, extracellular DNA and extracellular RNA (Fig. 2B) (45). Bacterial EPS

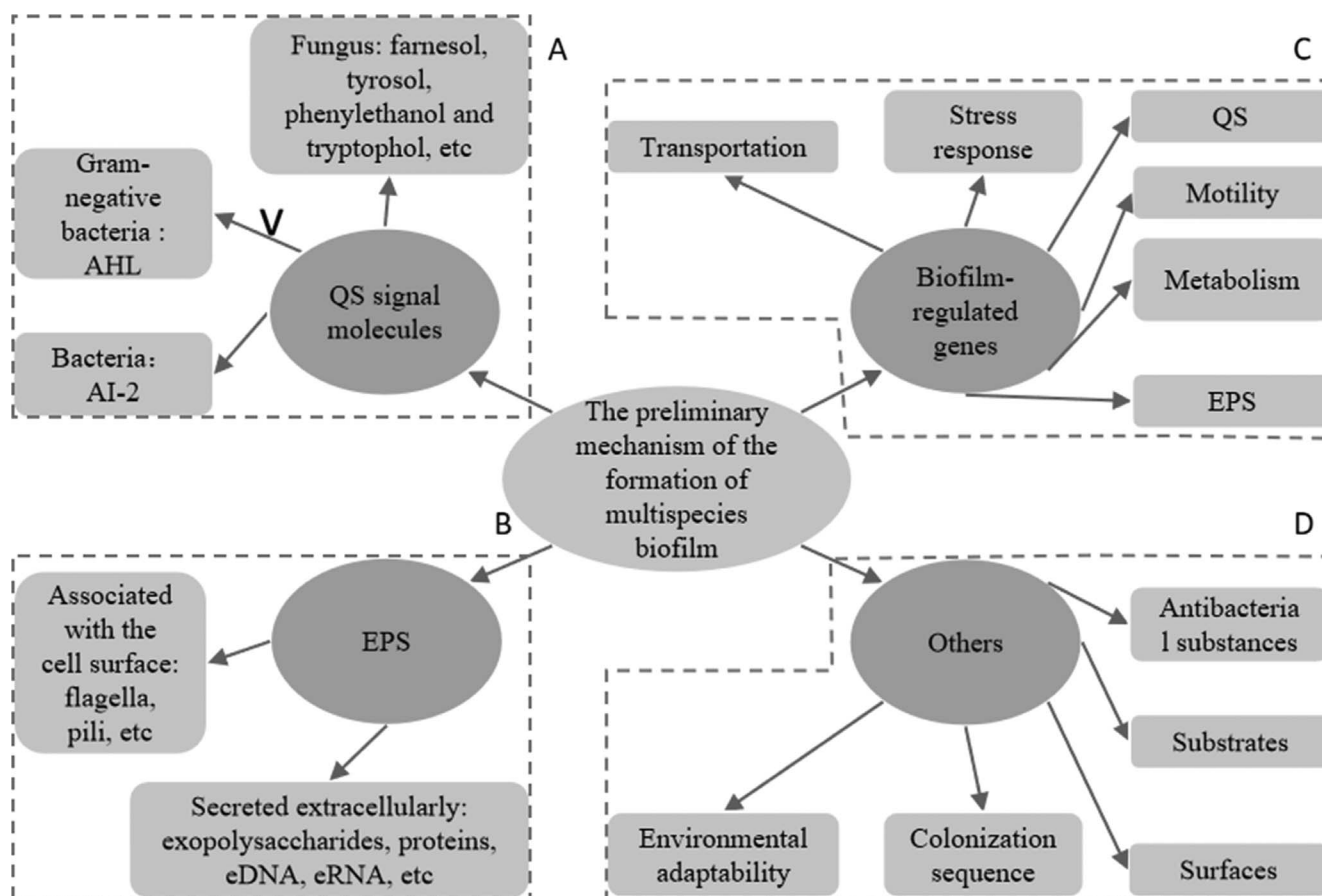


FIGURE 2. Preliminary mechanisms for the formation of multispecies biofilms: (A) QS signal molecules; (B) EPS; (C) biofilm-regulated genes; and (D) others.

components, such as curli fimbriae, cellulose, capsular polysaccharides, and lipopolysaccharides, have been reported to not only improve the biofilm formation ability of bacteria but also promote the viability and resistance of biofilms (104). The structure of the multispecies biofilms formed by *Salmonella* Typhimurium and *E. coli* O157:H7 strains is highly dependent on the characteristics of the coexisting strains, particularly EPS components, including curli fimbriae and extracellular polysaccharide cellulose (106). The bacteria that have EPS and fimbriae have an advantage over those types of bacteria that do not produce these components. However, those types of bacteria that lack these components can still form mixed biofilms by combining with types of bacteria that do have these features, which is likely to improve the persistence in the biofilm of both types of bacteria (73). Curli fimbriae endow *E. coli* O157:H7 with competitiveness in multispecies biofilms, possibly due to the interactions between Shiga toxin-producing *E. coli* and some spinach-associated microorganisms (10). The interaction between EPS protein TasA produced by *Bacillus subtilis* and extracellular polysaccharides produced by *Streptococcus mutans* plays a crucial role in the initial attachment of bacteria during the formation of dual-species biofilms (23). During the processing of ultrahigh-temperature-sterilized milk, the poor biofilm-forming bacterium *Lactococcus lactis* would benefit from the enhanced attachment provided by the

matrix of *P. fluorescens*, and in return, *P. fluorescens* would use the metabolites produced by *L. lactis* as a nutrient source (47). *L. plantarum* and yeast coexist in a wide range of fermented foods, and the ability of *L. plantarum* to adhere to surfaces with yeast cells enables the formation and maintenance of biofilms, and cell-to-cell attachment may occur through the interactions between a lectinlike protein of *L. plantarum* and mannan of yeast cells (37). *Pseudomonas* and *Basidiobolus* coexist in biofilms formed in the drinking water distribution system, and the extracellular enzymes secreted by the *Basidiobolus* can degrade high-molecular-weight compounds and release secondary metabolites, used by other microorganisms for growth (22).

The coexistence of different species or even different communities makes the analysis of the composition and biological characteristics of EPS more complex. Increasing details of EPS components in multispecies biofilms are being revealed; however, there is still very limited information about the functions and structures and how different microorganisms regulate the production and interaction of EPS components within biofilms. In addition to providing structural stabilities and functional environments, EPS also enhance the tolerance of biofilms to antimicrobials and disinfectants. Therefore, understanding the changes in the chemical composition of EPS in multispecies biofilms will help to exploit the weaknesses of pathogenic biofilms. Besides, an in-depth understanding

of the various properties of biofilm EPS can facilitate the development of effective methods for controlling biofilm-related contamination.

Biofilm-regulated genes. Regulatory genes are expressed at each stage of biofilm formation. The development of biofilms involves the regulation of hundreds of biofilm-specific genes, including those related to QS, EPS, metabolism, stress response, and transport (Fig. 2C) (21). Compared with the growth of individual species, some genetic characteristics affecting biofilm development will change, depending on the interaction among specific species (94). If there are genotypic differences in the expression of adhesin in multispecies biofilms, the strains with low adhesion may become increasingly dominant, because they can compete for more nutrients from highly adherent cells. As flagella genes of *L. monocytogenes*, *degU*, *flaA*, and *motB* play indispensable roles in controlling bacterial adhesion and regulating biofilm formation. *flaA* gene codes for flagellin, the principal component of bacterial flagellum, affecting the motility of strains (11). The MotB proteins, which form the stator component of the flagellum itself, serve a function in transmembrane transport of protons in the aforementioned flagellar motor complex (11). Strains with strong motility have weak adhesion, which is not conducive to the biofilm formation. DegU was shown to be essential for flagellar synthesis and bacterial motility in *L. monocytogenes*, and gradual increases in DegU phosphorylation levels are critical in the transition from motile to sessile biofilm-forming cells (33). Compared with single-species biofilms, these biofilm-related genes in the dual-species biofilms formed by *Vibrio parahaemolyticus* and *L. monocytogenes* were down-regulated, which may be a vital reason for the decreases in the biomass, biovolume, and thickness of the dual-species biofilms (13). During the biofilm development of *Trichoderma viride* and *Azotobacter chroococcum*, the attachment of *T. viride* to the mycelium would lead to significant changes in transcriptome, especially the expression of genes related to biofilm formation, including the genes encoding ATP-binding cassette (ABC) transporters, RNA-dependent RNA polymerase, translation elongation factor EF-1, dual homeobox 4, and molecular chaperones (100).

Horizontal gene transfer, the movement of genetic material between microorganisms, frequently occurs in microbial communities (44). Haubert et al. (36) demonstrated a transfer of the resistance gene *tetM* from *L. monocytogenes* to *E. faecalis* in Minas frescal cheese. Horizontal gene transfer can enable microorganisms to share genetic elements with other cells and is, therefore, associated with biofilm formation (63). Diani et al. (19) studied *Enterococcus faecium* and *E. faecalis* that are often detected in fermented sausages to understand the relationship between the presence of the *esp* gene and biofilm formation. As a result, the presence of the *esp* gene was found to promote the biofilm formation of *Enterococci*. Horizontal transfer of the *esp* gene may lead to the spread of biofilms, not only between *Enterococcus* strains, but also between different species, such as *Salmonella*. At present,

there are still very few reports on the promotion of biofilm formation by horizontal gene transfer; this is worthy of further research.

The formation of multispecies biofilms is related to the expression of regulatory genes of microorganisms in the biofilms, which is of great significance for understanding the adaptation and subsequent evolution of microbial characteristics. At present, there are still a number of gaps in these areas. In the future, more attention should be paid to these issues to improve our understanding of the mechanisms for multispecies biofilm formation.

Others. Environmental adaptability, colonization sequence, presence of antibacterial substances, substrates, or colonization surfaces of microorganisms also have great impact on the formation of multispecies biofilms (Fig. 2D). An increase in biofilm formation may be a common adaptive response for long-term coexistence. Enhanced biofilm formation is common for cocultured bacteria coexisting in the original environment. Reduction in biofilm formation will be even more pronounced when the cocultured bacteria are from environments without long-term coexistence of bacteria, indicating that the induction of interspecies biofilms is, at least partly, due to previous contacts and the formation of interspecies biofilms is an adaptive response (64). It is possible for microorganisms to recontact surfaces to form biofilms in the presence of preexisting biofilms formed by symbiotic or spoilage bacteria or other pathogens. Therefore, colonization sequence may play a key role in determining the composition and dominant species of multispecies biofilms. In the formation process of multispecies biofilms, early precolonization could significantly affect the competition between *E. coli* O157:H7 and *E. coli* O111:H8, and the growth of the precolonized colonies could exceed that of the latter colonized colonies (105). For example, after the precolonization of *L. monocytogenes*, the dual-species biofilms formed with *P. fluorescens* showed flat and thin-layer structures, while the precolonization of *P. fluorescens* would result in a multilayer structure with a higher thickness and roughness (83). Pang and Yuk (84) later found that colonization sequence does not affect populations of the dual-species biofilms of *L. monocytogenes* and *P. fluorescens* but influence the attachment of the dual-species biofilms. The result indicates that colonization sequence affects the structures and adhesion of the biofilms formed by *L. monocytogenes* and *P. fluorescens* but has no significant effect on cell density.

The formation mechanisms of multispecies biofilms vary among different surfaces or substrates. For different dairy substrates (brain heart infusion, cow's milk, and whey protein), *Enterococcus faecalis* was always dominant in multispecies biofilms, while other pathogens, including *L. monocytogenes*, *S. aureus*, and *Bacillus cereus*, could not form biofilms in a multispecies system with whey protein as the substrate, and only adhesion was observed (4). Iniguez-Moreno et al. (38) studied the formation of dual-species biofilms of *S. aureus* 4E and *Salmonella* and found that biofilms have better adhesion on stainless steel surfaces than

on polypropylene, which may be related to the improvement of iron use. Some microorganisms may release substances that inhibit biofilm formation. Some species of *Bacillus* are producers of antibacterial compounds (peptides, lipopeptides, bacteriocins, and inhibitory substances similar to bacteriocins) against bacteria and fungi (6). For example, *B. cereus* RC6 secretes two enzymes that degrade casein and show antibacterial activity against *L. monocytogenes* (77). *Staphylococcus xylosus* released a molecule with a weight greater than 30 kDa in its growth supernatant, which can resist proteolytic enzymes and inhibit the biofilm formation of *S. aureus* MW2 (56). *P. aeruginosa* inhibits the growth of *S. aureus* or can even inactivate *S. aureus* cells, due to the production of several toxic compounds, including hydrogen cyanide, pyocyanin, and alkyl-hydroxyquinoline *N*-oxide, which can block the electron transport chain of *S. aureus* (66).

Due to the complexity of microbial interactions, it is unreasonable to predict the formation of multispecies biofilms just on the basis of one single mechanism. The combination of QS, EPS, biofilm-regulated genes, environment, and other factors ultimately affects the behavior of microorganisms. For example, QS is highly correlated with EPS, which affects the function of biofilms. In the biofilms of *P. aeruginosa*, extracellular polysaccharides promote the absorption of QS signal molecules in the biofilms, and the biofilm matrix activates or inactivates the QS process (5). Compared with single-species biofilms, the dual-species biofilms formed by *V. parahaemolyticus* and *L. monocytogenes* exhibited significantly lower biomass and biological volume and thickness. These differences were ascribed to the number of bacteria in dual-species biofilms, expression of biofilm-regulated genes, metabolic activity, extracellular polysaccharides, and proteins (13).

MICROBIAL INTERACTIONS AMONG MULTISPECIES BIOFILMS

Interspecies interactions affect the biofilm phenotypes of bacteria. In recent years, there have been increasing studies of the interactions between major foodborne pathogens and other food-related microorganisms or microbial community members residing in food processing environments in multispecies biofilms. The interactions among multiple species are generally cooperative, competitive, or neutral and thus determine the dominant species in biofilms (Table 1).

Competitive interactions. In food processing environments, competitive interactions occur with one or more species at a disadvantage in coculturing. For instance, *P. aeruginosa* and *S. aureus* coexist in many biofilm-related infections and have a relationship that is competitive in nature (72). When cocultured with bacteria (*Citrobacter*, *Hafnia*, *Aeromonas*, and *Carnobacterium*) on meat processing surfaces, *Salmonella* Typhimurium and *E. coli* O157:H7 can be harmoniously integrated into competitive biofilms, while facultative anaerobes may have a competitive advantage over strict anaerobes in establishing multispecies biofilms (101).

The phenotypes involved in microbial competition in a multispecies biofilm may have developed to aid in competition for resources, such as space, nutrients, and energy through means of rapid and efficient growth, production of nutrient scavenging molecules, superior position in the niches, direct competition with other species by producing bacteriostatic substances, or through contact-dependent inhibition (46, 62, 88, 92, 96). *Vibrio cholerae* and *E. coli* have an antagonistic effect on the dual-species biofilms at the air-liquid interfaces, in which *V. cholerae* plays a dominant role in the viscous elasticity of the biofilms, and forms mature biofilms 18 h earlier than *E. coli* (1). In the multispecies biofilms containing *L. monocytogenes* or *S. aureus*, the number of sessile cells in each member of relevant microbial communities, including pathogens, is smaller than that in single- or dual-species biofilms, which may be ascribed to competition for nutrients (78). The competitive interaction between *Salmonella* and *P. aeruginosa* would result in a decrease in the density of dual-species biofilms on stainless steel coupons, which may be caused by AHL, a class of signal molecules produced by *P. aeruginosa* and reported to inhibit the *Salmonella* biofilm (81). The indole produced by *E. coli* can help resist the AHL-producing gram-negative bacteria and enhance its competitiveness in biofilms (89). The dual-species biofilms formed by *L. monocytogenes* and *B. cereus* isolated from dairy products exhibited competitive interactions as well, which is dependent on the ability of *B. cereus* to produce antibacterial metabolites against *L. monocytogenes* (3). However, competitive interactions are not always deleterious to microorganisms. For example, competition is beneficial when the coexisting species have overlapping metabolic niches (29). Moreover, competitive interactions enhance the resistance of dominant bacteria to disinfectants in multispecies biofilms, conducive to the survival of biofilms (49).

Cooperative interactions. Madsen et al. (64) proposed the notion of cooperative interacting microorganisms, considering that the total amount of multispecies biofilms was higher than the total amount of all single-species biofilms, and found that cooperative interactions account for 13% of the interactions among microorganisms in fresh water, ocean, and soil communities. Cooperative interactions may benefit single or multiple sides of microorganisms at the same time, which is conducive to the formation of biofilms. *Ralstonia insidiosa* and *Burkholderia caryophylli* isolated from fresh-cut processing facilities exhibited a synergy relationship with *E. coli*, and *E. coli* benefited from the cooperative interactions (60). A typical feature of cooperative interactions in microbial communities is that the partner usually excretes metabolites with a positive impact on the growth of the other microorganism (99, 112). Microorganisms may excrete metabolites through cell leakage or as by-products or end products of their own metabolism, which are absorbed by other microorganisms (57, 69). It has been reported that there were cooperative interactions in the growth of multispecies biofilms of *P. fluorescens*, *L. plantarum*, and *Leuconostoc pseudomesen-*

TABLE 1. Examples of microbial interactions among multispecies biofilms

Microorganisms in multispecies biofilms	Surfaces for biofilm formation	Types of interactions	Reference
<i>Salmonella</i> Typhimurium and meat processing surface bacteria (<i>Citrobacter</i> , <i>Hafnia</i> , <i>Aeromonas</i> , <i>Carnobacterium</i>)	Stainless steel coupons	Competitive	101
<i>Escherichia coli</i> O157:H7 and meat processing surface bacteria (<i>Citrobacter</i> , <i>Hafnia</i> , <i>Aeromonas</i> , <i>Carnobacterium</i>)	Stainless steel coupons	Competitive	101
<i>Vibrio cholerae</i> and <i>E. coli</i>	Air-liquid interface	Competitive	1
<i>Listeria monocytogenes</i> , <i>E. coli</i> , <i>Acinetobacter</i> , <i>Klebsiella</i> sp., and <i>Comamonas</i> sp.	Stainless steel coupons	Competitive	78
<i>Staphylococcus aureus</i> , <i>Aeromonas</i> , <i>Lactococcus lactis</i> , <i>Candida tropicalis</i> , and <i>Lactobacillus</i> sp.	Stainless steel coupons	Competitive	78
<i>Pseudomonas aeruginosa</i> and <i>Salmonella</i> spp.	Stainless steel coupons	Competitive	81
<i>E. coli</i> and gram-negative bacteria producing AHL	Silicon rubber disks	Competitive	89
<i>L. monocytogenes</i> and <i>B. cereus</i>	Stainless steel coupons	Competitive	3
<i>L. innocua</i> , <i>Rhodotorula mucilaginosa</i> , and <i>C. kefir</i>	Stainless steel coupons	Competitive	2
<i>L. monocytogenes</i> and <i>C. tropicalis</i>	Stainless steel coupons	Competitive	2
<i>Salmonella</i> Typhimurium and <i>Enterobacteriaceae</i>	96-well round bottom microtiter plates	Competitive	102
<i>Salmonella</i> Typhimurium and <i>Hafnia alvei</i>	96-well round bottom microtiter plates	Competitive	102
<i>Salmonella</i> Typhimurium and <i>Buttiauxella agrestis</i>	96-well round bottom microtiter plates	Competitive	102
<i>Salmonella</i> Typhimurium and <i>Serratia</i> sp.	96-well round bottom microtiter plates	Competitive	102
<i>E. coli</i> O157:H7 and <i>Ralstonia insidiosa</i>	Polystyrene 96-well microtiter plates	Cooperative	60
<i>E. coli</i> O157:H7 and <i>Burkholderia caryophylli</i>	Polystyrene 96-well microtiter plates	Cooperative	60
<i>P. fluorescens</i> , <i>Lactobacillus plantarum</i> , and <i>Leuconostoc pseudomesenteroides</i>	CDC Biofilm Reactors	Cooperative	54
<i>L. monocytogenes</i> and <i>R. insidiosa</i>	Glass slide surfaces	Cooperative	58
<i>Salmonella enterica</i> and <i>R. insidiosa</i>	Glass slide surfaces	Cooperative	58
<i>Listeria innocua</i> , <i>C. tropicalis</i> , and <i>Candida krusei</i>	Stainless steel coupons	Cooperative	2
<i>Salmonella</i> Typhimurium, <i>Carnobacterium maltaromaticum</i> , and <i>Buttiauxella noackiae</i>	96-well round bottom microtiter plates	Cooperative	102
<i>E. coli</i> and <i>L. monocytogenes</i>	Stainless steel American Iron and Steel (AISI) 304 #4 slides	Neutral	18
<i>A. hydrophila</i> and <i>Salmonella</i> Typhimurium	Crab coupons	Neutral	43
<i>A. hydrophila</i> and <i>L. monocytogenes</i>	Crab coupons	Neutral	43
<i>A. hydrophila</i> and <i>Pectobacterium carotovorum</i>	Crab coupons	Neutral	43

teroides, possibly because *P. fluorescens* produced extra EPS, which was absorbed by other microorganisms (54). The mixed culture of *L. monocytogenes* and *R. insidiosa* also displayed synergistic interactions, with the generation of a large amount of extra EPS in the dual-species biofilms, making the biofilms more robust than any single-species biofilms (58). Overall, cooperative interactions among microorganism in biofilms tend to promote adhesion, growth, and disinfectant resistance (3).

Other interactions. When multiple species coexist in the same biofilm environments, there may be a neutral state of noninterference with each other, which only happens in rare cases (109). For example, the presence of *E. coli* or its metabolites in the pretreatment medium did not affect the biofilm formation of *L. monocytogenes* (18). *Salmonella* Typhimurium, *L. monocytogenes*, and *Pectobacterium carotovorum* did not compete with *A. hydrophila*; instead, they showed neutral interactions with *A. hydrophila* in multispecies biofilms (43). Agustín and Brugnoli (2) simulated the interactions between *Listeria* and resident yeasts in the juice processing environment. As a result, the number of *L. innocua* cells increased significantly in the presence of *Candida tropicalis* and *Candida krusei* and

decreased significantly in the presence of *Rhodotorula mucilaginosa* and *Candida kefir*. When cocultured with *C. tropicalis*, *L. monocytogenes* also decreased significantly in the number of cells. These results reveal the synergistic or antagonistic interactions between *Listeria* and resident yeasts; therefore, cooperative and competitive interactions may exist simultaneously among multiple species.

The interactions among microorganisms in multispecies biofilms are affected by the environment and their own properties. Competitive interactions are more common than cooperative and neutral interactions under conditions of limited nutrition and space (41). The cooperative and competitive interactions will largely determine the dominant species in the mixed biofilms, thus determining the structure and activity of the multispecies community. Therefore, in-depth exploration of microbial interactions in multispecies biofilms is highly necessary for understanding the interactions in multispecies biofilms and how these interactions affect biofilm resistance to disinfectants, which may help avoid the formation of harmful microbial biofilms and promote that of desired microbial biofilms.

TABLE 2. Examples of multispecies biofilms with enhanced disinfectant resistance

Microorganisms in multispecies biofilms	Disinfectants	Surfaces for biofilm formation	Reference
<i>Pseudomonas aeruginosa</i> and <i>Listeria innocua</i>	QAC, tertiary alkyl amine, and chlorine based	Polycarbonate coupons	49
<i>Staphylococcus aureus</i> and <i>L. innocua</i>	QAC, tertiary alkyl amine, and chlorine based	Polycarbonate coupons	49
<i>Listeria monocytogenes</i> and <i>P. fluorescens</i>	Benzalkonium chloride	Stainless steel coupons	83
<i>Escherichia coli</i> and <i>Enterococcus faecalis</i>	Hydrogen peroxide	8-well glass bottom slides	51
<i>Staphylococcus aureus</i> , <i>Aeromonas</i> spp., <i>Lactococcus lactis</i> , <i>C. tropicalis</i> , and <i>Lactobacillus</i> sp.	Peracetic acid	Stainless steel coupons	78
<i>P. aeruginosa</i> and <i>Salmonella</i> spp.	Chlorine and Ecolab Whisper V ^a	Stainless steel coupons	81
<i>Salmonella</i> Enteritidis and <i>P. fluorescens</i>	QAC	Stainless steel coupons	79
<i>P. fluorescens</i> and <i>S. aureus</i>	Carvacrol	Stainless steel coupons	107
<i>Salmonella</i> Enteritidis and <i>P. aeruginosa</i>	Chlorine	Stainless steel coupons	82
<i>P. fluorescens</i> and <i>L. monocytogenes</i>	Benzalkonium chloride	Stainless steel coupons	84
<i>Lactobacillus brevis</i> , <i>Lactobacillus lindneri</i> , and <i>Pediococcus pentosaceus</i>	Peracetic acid	Stainless steel coupons	67
Cultivable indigenous microorganisms from fresh salmon and <i>L. monocytogenes</i>	Ecolab Whisper V	Stainless steel coupons	80
Species isolated from brewery contaminated biofilms (<i>Pseudomonas</i> , <i>Raoultella</i>)	Sulfathiazole	96-well plates	86
<i>Pseudomonas libanensis</i> and <i>A. hydrophila</i>	Hydrogen peroxide, peracetic acid and sodium hypochlorite	Stainless steel coupons	111
<i>P. libanensis</i> and <i>Chryseobacterium oncorhynchi</i>	Hydrogen peroxide, peracetic acid, and sodium hypochlorite	Stainless steel coupons	111

^a Ecolab Whisper V: a blend of four effective QACs.

PRELIMINARY MECHANISMS FOR THE RESISTANCE OF MULTISPECIES BIOFILMS TO DISINFECTANTS

Increasing evidence has indicated that the interactions among different species can greatly increase the resistance of multispecies biofilms to biocides (95). Compared with single-species biofilms, multispecies biofilms are characterized by greater resistance to disinfectants, such as chlorine, benzalkonium chloride, sodium hypochlorite, peracetic acid, quaternary ammonium compounds (QAC), and hydrogen peroxide (Table 2). The studies concerning the resistance of major foodborne pathogens to some industrial disinfectants have also revealed that disinfectants cannot completely inactivate target microorganisms in multispecies biofilms (14, 26). However, the mechanisms underlying the resistance of multispecies biofilms are not fully understood yet. However, some factors potentially associated with the increase in resistance of microorganisms to disinfectants in multispecies biofilms have been proposed, such as the type of strains, EPS formation, biofilm age, environment, and spatial distribution of strains.

Type of strains. In most cases, multispecies biofilms are more resistant to disinfection than single-species biofilms, and dual-species biofilms with foodborne pathogens are more resistant to disinfection treatment than single-species biofilms, as well (49). For example, the presence of *P. fluorescens* in the dual-species biofilms could protect *L. monocytogenes* cells against disinfectant treatments (83).

When *E. coli* and *E. faecalis* were cocultured to form biofilms, the disinfectant resistance of both species was enhanced, and the coaggregation in the biofilms also greatly improved the survival rate of *E. faecalis*, because the *E. faecalis* cells in the biofilms were covered by small colonies of *E. coli* (51). Oxaran et al. (78) established mature multispecies biofilms containing *L. monocytogenes* and *Staphylococcus aureus* on stainless steel coupons and treated them with peracetic acid. As a result, *S. aureus* in multispecies biofilms has a bigger number of sessile cells than that in single-species biofilms, indicating that some specific microorganisms in multispecies biofilms can provide certain protection on other microorganisms. This may be due to the presence of specific microorganisms that affect the composition of the biofilm matrix and are more resistant to disinfectants.

However, the interactions among different strains in multispecies biofilms may also have no positive impact on the disinfection resistance of microorganisms. Gomes et al. (31) isolated two pathogens from drinking water, namely, *Acinetobacter calcoaceticus* and *Stenotrophomonas maltophilia*, and found that the dual-species biofilms formed by the two microorganisms on polyvinyl chloride surfaces were more sensitive to sodium hypochlorite than single-species biofilms. In the dual-species biofilms formed by *Salmonella* Typhimurium and *S. aureus*, Gkana et al. (30) found that the interspecies interactions had no significant effect on the overall disinfectant tolerance (benzalkonium chloride, sodium hypochlorite, and peracetic acid) of the

whole species. In addition, Iniguez-Moreno et al. (38) found that the interspecies interactions between *S. aureus* 4E and *Salmonella* had a negative effect on the antimicrobial resistance of each microorganism, compared with the single-species biofilms. These findings suggest that the increase in the resistance of multispecies biofilms to disinfectants is not a universal phenomenon.

EPS. As a physical barrier, EPS limit the penetration of disinfectants or the interactions of microorganism with disinfectants, thus hindering the function of disinfectants (9). In multispecies biofilms, EPS produced by one species may protect another species from biocides by limiting the diffusion process (95). For example, previous studies have demonstrated the critical role of EPS in protecting the multispecies biofilms of *Salmonella* Typhimurium and culturable native microorganisms in lettuce from UV-C radiation (42). *P. aeruginosa* was found to inhibit the growth of *Salmonella* cells in biofilms but protect *Salmonella* cells in biofilms from chlorine or Ecolab Whisper V by providing more EPS in dual-species biofilms than in single-species biofilms (81). *P. fluorescens* could produce more polysaccharides in dual-species biofilms, which have a protective effect on *Salmonella* Enteritidis, thereby enhancing the resistance of *Salmonella* Enteritidis to QAC (79). The interaction between *P. fluorescens* and *S. aureus* in the dual-species biofilms could promote the production of extracellular polysaccharides, which may contribute to an increase in the resistance of the communities to carvacrol (107). In the dual-species biofilms formed on stainless steel coupons in chicken juice, *Salmonella* Enteritidis showed higher chlorine resistance, which may be due to the presence of more glycoconjugates in the dual-species biofilms to hinder the effect of chlorine (82).

Biofilm age. The signals and metabolites may be different between the microorganisms in biofilms and the planktonic state, which may cause different physiological changes in neighboring cells. Therefore, the resistance of microorganisms to disinfectants in multispecies biofilms is related to the biofilm age as well. Mature biofilms are generally more tolerant to stress conditions than freshly formed biofilms due to the strong three-dimensional structure of adherent layers of bacterial cells, which constitute a physical barrier that restricts and blocks the penetration of disinfectants or other chemicals (104). For example, the formation of biofilms with preexisting *P. fluorescens* on stainless steel enhanced the resistance of *L. monocytogenes* to disinfectants, though the setting reduced the transfer rate of *L. monocytogenes* to salmon slices (84). Most of the spoilage bacteria in the microbrewery environment showed strong biofilm-forming ability on polystyrene and stainless steel, while the mature biofilm on stainless steel was not sufficiently inactivated by peracetic acid (67). Therefore, in food processing environments, cleaning and sanitation of equipment and utensils should be carried out as early as possible, helping reduce the residue of multispecies biofilms.

Other factors. In addition to the types of strains, EPS, and biofilm ages, there are other factors related to the disinfectant resistance of multispecies biofilms, such as environment and spatial distribution of strains. In terms of the environment, the surface shape can greatly affect the adhesion of microorganisms, especially when the surface has channels to protect the microorganisms from disinfectants. Depending on the physiological states of the bacteria and the molecules present during exposure, nutritional status can also alter the antibiotic sensitivity of multispecies biofilms (40). For example, the addition of CO₂ could antagonize the effect of antibiotics on multispecies biofilms (39). Due to the specific spatial distribution of some species within the biofilms, some strains may be protected from disinfectants through aggregation with different three-dimensional structures from other strains (55). *E. coli* O157:H7 mainly appeared in the bottom layer of the dual-species biofilm, which provided *R. insidiosa* with better protection, including resistance to disinfection treatments (59). The cultivable local microorganisms from fresh salmon could form multispecies biofilms with *L. monocytogenes*, which had stronger antidisinfectant capacity than single-species biofilms, possibly because the honeycomb cell cluster shielded the underlying structures of the biofilms (80).

Overall, the enhanced resistance of multispecies biofilms to disinfectants may not depend on only one single factor but on a combination of multiple factors. For example, the biofilm age affects the production of EPS, which will, together, influence the disinfectant resistance of biofilms. Moreover, it is not feasible to predict the resistance of multispecies biofilms to disinfectants on the basis of the impact of a single species; this needs to be considered comprehensively according to the actual situation.

CONCLUSIONS AND FUTURE PERSPECTIVES

Interactions among microorganisms occur frequently and may eventually lead to the formation of dense, complex, and highly structured multispecies biofilms during food processing. The cooperative, competitive or neutral interactions between foodborne pathogens (such as *Salmonella*, *L. monocytogenes*, *S. aureus*, and *E. coli*) and other microorganisms in the biofilms may enhance pathogenicity. Understanding the preliminary mechanisms for the formation of multispecies biofilms can provide directions for blocking and eliminating the formation of some undesired biofilms. Using the antagonistic role of other microbial species may contribute to the development of methods for controlling pathogenic bacteria in food processing environments. However, the increased disinfectant resistance of most multispecies biofilms remains a great challenge in the food industry. In this article, the preliminary mechanisms for the resistance of multispecies biofilms to disinfectants were discussed, such as strain type, EPS, biofilm age, and nutritional status. Further exploration of these factors may help develop new and more efficient disinfectants. In addition, the combination of disinfectants with other methods (such as enzymes, bacteriophages, ultrasound,

and ozone) may greatly improve the control efficiency on multispecies biofilms.

At present, the mechanism underlying the formation of multispecies biofilms is still not fully understood. Many laboratory-based studies of multispecies biofilms have been focused on low-diversity biofilms, such as biofilms consisting of two to four species. With the increase in cohabitating microorganisms, there will be a significant increase in the population number, which will increase the complexity of research. Although such studies may be difficult, it is a good way to explore the formation mechanisms of biofilms and the mechanisms for the disinfectant resistance mechanisms of multispecies biofilms with more than four species by using high-throughput sequencing techniques, such as transcriptomics, proteomics, and metabolomics. There are already many studies that have used confocal laser scanning microscopy and have three-dimensional renderings of multispecies biofilms. In the future, the dynamic process of multispecies biofilm formation can be explored by combining fluorescence in situ hybridization and confocal laser scanning microscopy and finally rendering three-dimensional visualization of the formation process of biofilms. Further studies of these issues will help find suitable and efficient ways to prevent and control the formation of multispecies biofilms in food processing environments.

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